Optics became clearer — Solution

X26 (2026) — English translation

A1 **0.60 pts**

For 5 different values of a, obtain 5 corresponding values of h at which the LED image S' is focused on the screen. For each measurement, calculate the focal length value f. Average the resulting values f and write down the average value f_{avg} on your answer sheet.

a, mm b, mm f, mm 210 828 167.5 225 694 169.9 235 620 170.4 252 528 170.5 280 429 169.4

Answer: Answer:

$$f_{\text{avg}} = (170 \pm 2)\ \text{mm}$$

B0 **0.10 pts**

Measure and write down the distance a_0 on your answer sheets.

Answer: Answer:

$$a_0 = (207 \pm 2)\ \text{mm}$$

B1 **0.40 pts**

By changing the position of the d aperture, observe the changes in the image on the screen. Mark (✓) all correct statements on your answer sheets.

As the aperture approaches the source, the brightness of the image increases; as the aperture moves away from the source, its brightness decreases.

Answer: Answer: The size of the image and its shape remain unchanged. As the aperture approaches the source, the brightness of the image increases; as the aperture moves away from the source, its brightness decreases.

B2 **0.60 pts**

For 8 different distances d between the source and aperture D_2, measure the dimensions m_1', m_2' of both diagonals of the cross image on the screen. Vary d as much as possible. Calculate for each d the average image size m'.

d, mm $m1'$, mm $m2'$, mm $1/d$, m^{-1} m', mm 50 88 88 20.0±0.80 88 75 88 88 13.3±0.36 88 100 88 88 10.0±0.20 88 125 88 88 8.00±0.13 88 143 85 85 6.99±0.10 85 160 64 64 6.25±0.08 64 182 55 55 5.49±0.06 55 204 49 49 4.90±0.05 49 226 43 43 4.42±0.04 43 244 40 40 4.10±0.03 40 262 35 35 3.82±0.03 35 287 32 32 3.48±0.02 32 321 28 28 3.12±0.02 28 350 27 27 2.86±0.02 27 ±2 ±1 ±1 ±1

B3 **1.50 pts**

Obtain the theoretical dependence $m'(d)$ via m, d, f, a_0. Plot a linearized graph and obtain the size m of the cross.

Let us highlight with a dotted line a plane at a distance b from the source, which is focused by the lens onto the screen. It is obvious that from this plane only rays limited by the dotted line will reach the screen. Then we can consider an image of size m' on the screen as an image of a segment of length l in the indicated plane.

$$l = \frac{m}{d} \cdot b$$

Then size image on screen:

$$m' = l \cdot \frac{h - a_0 - f}{a_0 + f - b}$$

Formula thin lens:

$$\frac{1}{f} = \frac{1}{a_0 + f - b} + \frac{1}{h - a_0 - f}$$

From which:

$$a_0 + f - b = f \cdot \frac{h - a_0 - f}{h - a_0 - 2f}$$

Исключая b obtain:

$$m' = -\frac{m}{fd}(h - a_0 - 2f) \cdot (f \cdot \frac{h - a_0 - f}{h - a_0 - 2f} - a_0 - f)$$

13Substitute $h - a_0$ from the thin lens formula:

$$h - a_0 = \frac{a_0 f}{a_0 - f}$$

$$m' = \frac{m}{d} \cdot \left(\frac{a_0}{a_0 - f}(2f - a_0) - f \right) = \frac{m}{d} \cdot \left(\frac{a_0 f}{a_0 - f} - (a_0 + f) \right)$$

Линеаризация:

$$m'(1/d)$$

Dots which lie on the plateau - aperture by the lens; we will not plot them on the graph, because they are not needed to calculate the slope factor.

From the resulting graph we learn the slope coefficient of the line:

$$k_1 = (1.21 \pm 0.06) \cdot 10^{-2} \text{ m}^2$$

Express m ч:

$$m = \frac{k_1}{\frac{a_0 f}{a_0 - f} - (a_0 + f)}$$

Итого:

$$m = 21 \text{ mm}$$

$$\Delta m = 3 \text{ mm}$$

$$k_1 = (1.21 \pm 0.06) \cdot 10^{-2} \text{ m}^2$$

Let's express m h:

$$m = \frac{k_1}{\frac{a_0 f}{a_0 - f} - (a_0 + f)}$$

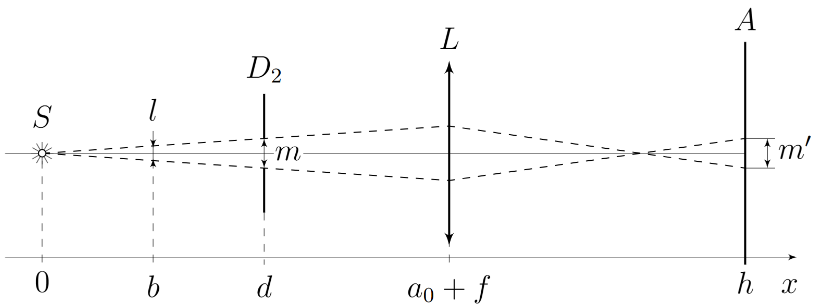
Total:

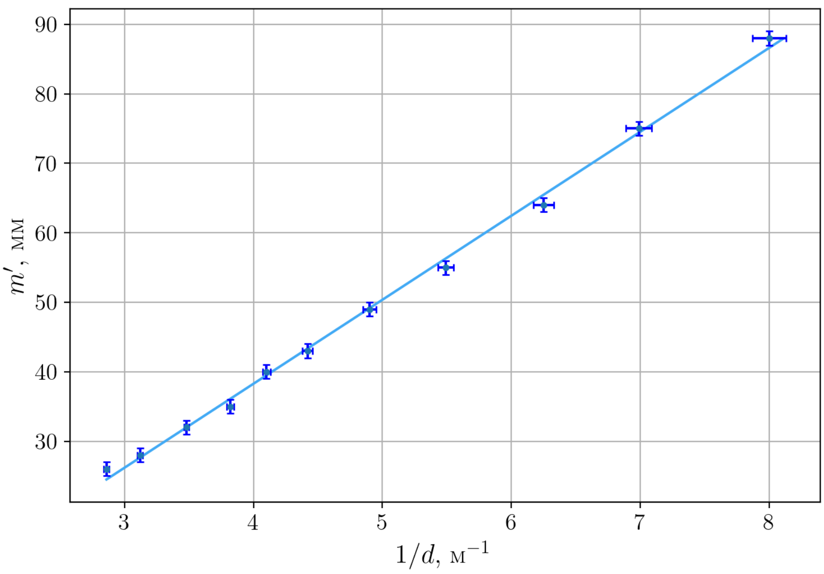
$$m = 21 \text{ mm}$$

$$\Delta m = 3 \text{ mm}$$

Answer: Answer:

$$m = (21 \pm 3) \text{ mm}$$



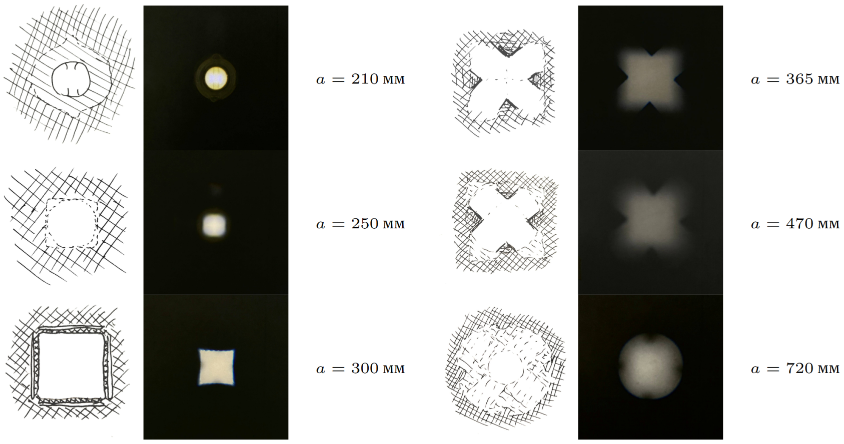


C1 — 1.80 pts

As you move the lens, the appearance of the image on the screen will change. Vary the position of the lens (i.e. the value of a) and for each a sketch the appearance of the image on your answer sheets. Your image should clearly indicate (by shading or otherwise) highlight, shadow, and penumbra (if any).

Commentary on pictures. $a = 210$ mm - focused LED; $a = 250$ mm - blurry circle (almost a square); $a = 300$ mm - focused square (stripes along the edges are reflections from the edges); $a = 365$ mm - cross, almost focused, cut into a square; $a = 470$ mm - cross, cut into a square, blurred; $a = 720$ mm - a cross cut off by a circle (lens), very blurred.

Commentary on pictures.



C2

0.90 pts

Measure and write down a_1. Changing only the position of the aperture D_2 (cross), take 8 points depending on the minimum horizontal width v_2 of the image (the width at the narrowest point) on the distance d_2 between the source and the aperture D_2. Vary d_2 as much as possible.

$$a_1 = 295 \text{ mm}$$

d_2, mm v_2, mm $1/d_2$ m^{-1} 132 20 7.58±0.11 155 20 6.45±0.08 175 18.5 5.71±0.07 192 16.5 5.21±0.05 203 16 4.93±0.05 215 15 4.65±0.04 227 14 4.41±0.04 245 13.5 4.08±0.03 265 13 3.77±0.03 ±2 ±0.5

C3

0.90 pts

Measure and write down a_2. Changing only the position of the aperture D_1 (square), take 8 points depending on the maximum diagonal width v_1 (along the straight line AB shown in the figure to point B2) of the image on the distance d_1 between the source and the diaphragm D_1. Vary d_1 as much as possible.

$$a_2 = 382 \text{ mm}$$

d_1, mm v_1, mm $1/d_1$, m^{-1} 22 63 45.4±4.13 41 63 24.3±1.19 53 63 18.8±0.71 68 63 14.7±0.43 73 59 13.7±0.38 88 49 11.3±0.26 97 43 10.3±0.21 106 39 9.43±0.18 114 34 8.77±0.15 128 28 7.81±0.12 ±2 ±2

C4

0.80 pts

Let the distance a between the lens and the source be such that a focused image of the aperture D_1' is obtained on the screen. Find the dependence of the image size l_2 on the screen on the distance d_2 between the source and the second aperture D_2'. Find d_2', at which the nature of the dependence changes. Express your answers using h, a, d_1, d_2, p_1, p_2.

It is clear from the figure that for small d_2 the image will look like the distribution of light intensity in the section D_1, and will not change when d_2 changes, up to a certain value ($d_2 = d_2'$).

Starting from d_2', some of the rays passing through the diaphragm D_1 will not pass through the diaphragm D_2. This moment occurs when the angular dimensions of the diaphragms are the same. Then:

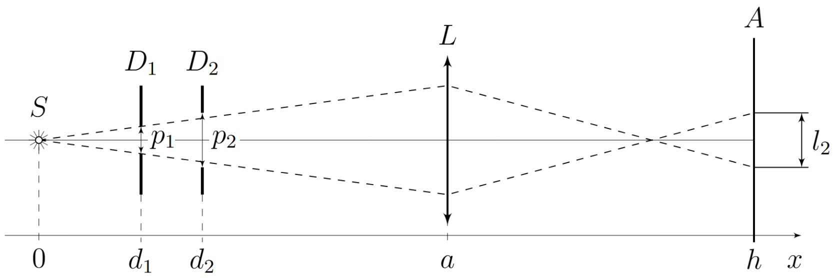
$$d_2' = \frac{p_2}{p_1} \cdot d_1.$$

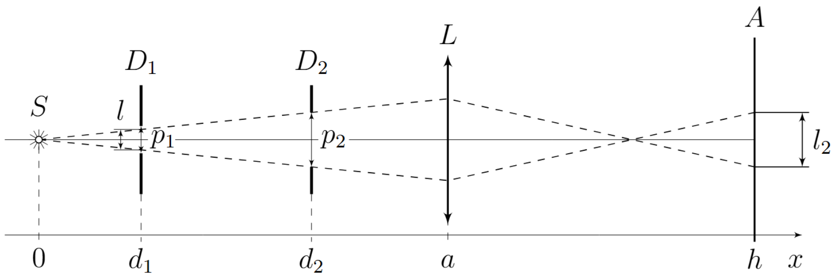
Now find l_2. Similarly part B3 can consider segment l in plane D_1 ограниченный крайнии ray pass through through diaphragm D_2 (or D_1, if $d_2 d_2' = \frac{p_2}{p_1} \cdot d_1 . Nowlet's find l_2$. Similar to point B3, we can consider the segment l in the plane D_1 limited by the extreme rays passing through the diaphragm D_2 (or D_1, if $d_2 l_2 = \frac{h-a}{a-d_1} \cdot l, Where : l = \begin{cases} p_1, & d_2 < d_2' \\ \frac{d_1}{d_2} \cdot p_2, & d_2 > d_2' \end{cases}$ **Answer:**$d_2' = \frac{p_2}{p_1} \cdot d_1, l_2 = \begin{cases} \frac{h-a}{a-d_1} \cdot p_1, & \text{for } d_2 < d_2' \\ \frac{h-a}{a-d_1} \cdot \frac{d_1}{d_2} \cdot p_2, & \text{for } d_2 > d_2'. \end{cases}$

Answer: Answer:

$$d_2' = \frac{p_2}{p_1} \cdot d_1,$$

$$l_2 = \begin{cases} \frac{h-a}{a-d_1} \cdot p_1, & \text{for } d_2 < d_2' \\ \frac{h-a}{a-d_1} \cdot \frac{d_1}{d_2} \cdot p_2, & \text{for } d_2 > d_2'. \end{cases}$$





C5 — 0.40 pts

Let now the distance a between the lens and the source be such that a focused image of the diaphragm D_2' is obtained on the screen. Find the dependence of the image size l_1 on the screen on the distance d_1 between the source and the second aperture D_1'. Find d_1', at which the nature of the dependence changes. Express your answers using $h,\ a,\ d_1,\ d_2,\ p_1,\ p_2$.

Having written equations similar to C4, we get the answer:

Answer: Answer:

$$d_1' = \frac{p_1}{p_2} \cdot d_2$$

$$l_1 = \begin{cases} \frac{h-a}{a-d_2} \cdot p_2, & \text{for } d_1 < d_1' \\ \frac{h-a}{a-d_2} \cdot \frac{d_2}{d_1} \cdot p_1, & \text{for } d_1 > d_1'. \end{cases}$$

C6 — 0.60 pts

Obtain expressions for $v_1(d_1)$ and $v_2(d_2)$. Express your answers using $h,\ a,\ d_1,\ d_2,\ m,\ n,\ c$.

Using the result of the previous paragraph for point 2 we can match

$$\begin{cases} p_1 = c \\ p_2 = n\sqrt{2} \\ l_2 = v_2 \\ a = a_1 \end{cases}$$

then we get:

For C3:

$$\begin{cases} p_1 = c\sqrt{2} \\ p_2 = m \\ l_1 = v_1 \\ a = a_2 \end{cases}$$

Total we get:

Answer: Answer:

$$d_2' = \frac{n\sqrt{2}}{c} \cdot d_1,$$

$$v_2 = \begin{cases} \frac{h-a_1}{a_1-d_1} \cdot c, & \text{for } \ d_2 < d_2' \\ \frac{h-a_1}{a_1-d_1} \cdot \frac{d_1}{d_2} \cdot n\sqrt{2}, & \text{for } \ d_2 > d_2'. \end{cases}$$

C7 0.70 pts

Plot a linearized graph of the relationship taken in step C2 and obtain the sizes of c and n apertures. An error estimate for the final response c and n is not required.

Linearization - $v_2(1/d_2)$.

From the graph we find the slope coefficient, the size of the image on the plateau, and the break point:

$$k_2 = 2.78 \cdot 10^{-3}\ \mathrm{m}^2 v_2' = 20\ \mathrm{mm} d_2' = 156\ \mathrm{mm}$$

Answer: Answer: From the straight line slope coefficient:

$$n = \frac{k_2(a_1 - d_1)}{\sqrt{2}\ d_1(h - a_1)} = 5.7\ \mathrm{mm}$$

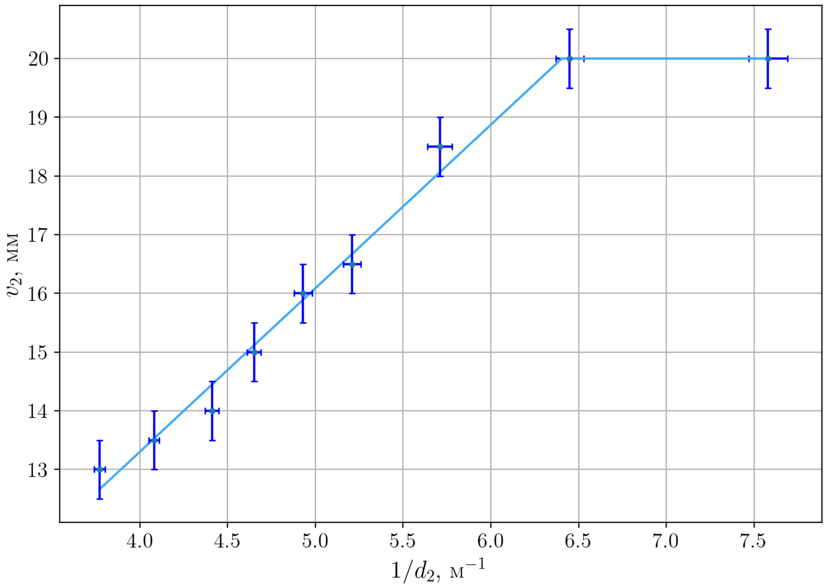
From value плато:

$$c = \frac{v_2'(a_1 - d_1)}{h - a_1} = 4.9\ \mathrm{mm}$$

From point излома and value плато:

$$n = \frac{c\ d_2'}{\sqrt{2}\ d_1} = 6.4\ \mathrm{mm}$$

Straight measurement $n = 6.3\ mm,\ c = 5.3\ mm$.



C8 — 0.70 pts

Plot a linearized graph of the relationship taken in step C3 and obtain the sizes of c and m apertures. An error estimate for the final response c and m is not required.

Linearization - $v_1(1/d_1)$

From the graph we find the slope coefficient, the size of the image on the plateau, and the break point:

$$k_3 = 4.99 \cdot 10^{-3}\ \text{m}^2 v_1' = 63\ \text{mm} d_1' = 69\ \text{mm}$$

Answer: Answer: From the straight line slope coefficient:

$$c = \frac{k_3(a_2 - d_2)}{\sqrt{2}\ d_2(h - a_2)} = 5.7\ \text{mm}$$

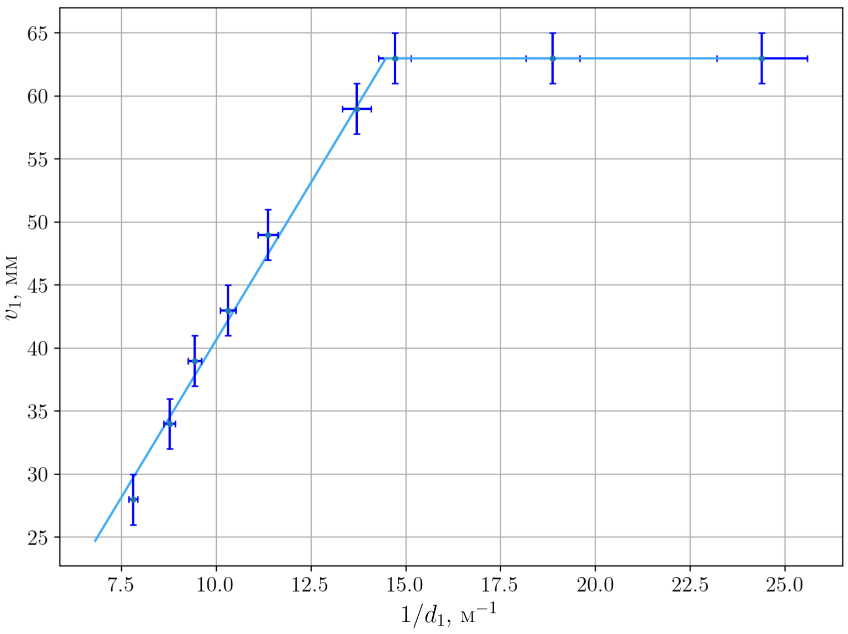
From value плато:

$$m = \frac{v_1'(a_2 - d_2)}{h - a_2} = 17.4\ \text{mm}$$

From point излома and value плато:

$$m = \frac{\sqrt{2}\ c\ d_2}{d_1'} = 19.9\ \text{mm}$$

Straight measurement $m = 17.3\ mm,\ c = 5.3\ mm$.


v_1, мм
$1/d_1$, м$^{-1}$
65
60
55
50
45
40
35
30
25
7.5
10.0
12.5
15.0
17.5
20.0
22.5
25.0